Roll No. | | | | | | | | | | | | | **Total No. of Pages : 02**

Total No. of Questions : 13

B.Pharmacy (Sem.-8)

CELL AND MOLECULAR BIOLOGY

Subject Code : BP-808ET

M.Code : 79771

Date of Examination : 13-01-2023

Time : 3 Hrs. **Max. Marks : 75**

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.**

SECTION-A

1. Write briefly :

a) Which type of cells possesses smooth ER?

b) Name two types of ribosomes.

c) What is mutation?

d) Define the terms "*Transcription*" and "*Translation*".

e) What is genotype and phenotype?

f) What is repression?

g) Name the chemical components of a ribosome.

h) What are lysosomes? Name different types of lysosomes.

i) What is reverse transcription?

j) What are secondary messengers? Give two examples.

SECTION-B

2. Define Cell and Molecular biology. Discuss in detail about applications of molecular biology.

3. Describe the various phases involved in the mitotic division of an animal cell.

4. Mention various types of receptors for cell signaling. Discuss in detailed about –G-protein coupled receptors.

SECTION-C

5. Write a note on cell membrane.

6. Describe briefly the process of transcription and translation.

7. Describe different types of cell cycle check points and their significance.

8. Describe process of transgenics with special reference to transgenic animals.

9. Describe in detail misregulation of signaling pathway.

10. Explain briefly about "*Electron transport system*".

11. Add a note on different types of protein structures.

12. Write a note on AMPK signaling.

13. Add a note on Tyrosine-specific protein kinases.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.